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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 320 — Past Questions & Materials

by Renance

Course: EEE 320

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 200, 300

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Paper 1 of 3 — EEE 320 - 2024/2025 - Second Semester - Measurement and Instrumentation

Level 200 • Second Semester • PastQuestion

SECTION B

QUESTION 4

State the three (3) various types of transducers with one example each, and their respective working

Principles (6 marks)

b. State the formula of a parallel plate capacitor and define its parameters (2 marks)

c. Differentiate between analogue and digital voltmeters (2 marks)

d. A parallel plate capacitor kept in the air has an area of 0.50012 m^2 and is separated from each other by a distance of 0.04 m , Calculate the parallel plate capacitor

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Paper 2 of 3 — EEE 320 MEASUREMENT AND INSTRUMENTATION PQ

Level 300 • Second Semester • PastQuestion

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TRONICS ENGINEERING

FEDERAL UNIVERSITY OF PETR

COLLEGE OF ENGINEERI

DEPARTMENT OF ELECTRICAL AND ELEC

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EXAMINATION: 2nd Semester a SESSION: 2023/2024

COURSE CODE: EEE 320 SEGRRE BENG:

TIME ALLOWED: 3 Hours

DATE: 25th Sept. 2024

Start each answer on afresh page

COURSE TITLE: Measurement and
Instrumentation

uestions.

INSTRUCTION: Answer any 5

QUESTION 1

coil (PMMC)

a. With a suitable sketch.

D°Arsonval meter (6 marks)

b._ A series circuit of two resistors $R_a = 25k\Omega$ and $R_b = 10k\Omega$ is supplied by a 30V supply. A voltmeter 1s

used to measure V_{pp} . The sensitivity 0 meter is used on 10V range. Draw

the circuit and Find the percentage error (5 marks)

Define the following measurement terms:

(3 marks)

describe the principle of operation of a permanent magnet moving

f the voltmeter is $1k\Omega/V$. the

due to its loading effect.

c.

(i) Resolution (ii) Sensitivity (iii) Reproducibility

QUESTION 2

a. With the aid of necessary circuit diagram; derive the general formula for the balance condition of an

AC bridge circuit. sae (6 marks)

'b. The impedances of an AC bridge is given as: $Z_1 = 200 \angle 30^\circ \Omega$, $Z_2 = 150 \angle 0^\circ \Omega$, $Z_3 = 250 \angle -40^\circ \Omega$.

Determine the constant of the unknown arm. (4. marks)

c. Explain the different types of forces necessary for satisfactory operation of Electromechanical indicating instrument. . (4 marks)

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QUESTION 3 Ake ieee

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a. Draw Maxwell bridge circuit and derive the formula for R_x and L_x 4 be sutaed th a ree marks)

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ae ttentns ont og (© marks)

b. State the laws of thermocouple

d. Using the Schering bridge in figure 1 below, calculate the values of the
fe ssn ends C_y and the equivalent series resistance R_x i] Laat Oe teenie (3 marks)
ower factor and Joss angle a Wea d lace 90° .

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$R_Z = 300 \text{ ghms } \% \text{ dhe b wie}$

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QUESTION 4

a. With a suitable sketch, describe the operating principle of a Cathode ray tube (CRT) Oscilloscope
. i " © (6 marks}

b. State any four precautions taken using multimeter for measurement (4 marks)

——""With the aid of a circuit diagram, design a multi-range ammeter whose ranges are 1mA, 10mA
and

100mA. Using a 50#A D*Asonval movement with a resistant of 200022. (4 marks)

QUESTION 5

a. With a neat sketch, explain the operation of a Linear Variable Differential Transformer (LVDT)
and
(8 marks)

state its applications

- b. Outline the various tests use for locating underground cable faults (4 marks) dy
- c. A parallel plate capacitor kept in the air-has an area of 0.50m^2 and is separated from each other by a distance of 0.04m . Calculate the capacitance of the parallel plate? (2 marks)

QUESTION 6 bs

- a. .With a simple sketch, describe the operation of a strain gauge. (6 marks)
- b.. Briefly explain.the types-of faults that.are most likely to occur with underground cables (6 marks)
- 'ce. .Calculate the output voltage of a 4- ve ud 2R ladder DAC with a reference voltage $V_{\text{rer}} = 5\text{V}$ and input binary code 1011 Ceyne ey (2 marks)

QUESTION 7

- a. i. What is thermistor? State the ailvanitages of thermistors (5 marks)
- ii. Briefly describe the following and state their applications
NTC, PTC, RTD, ADC and DAC i (5 marks)
- b. A strain gauge' has a gauge factor of 2. The original resistance is 120Ω . If a strain of 0.001 is applied, calculate the change in resistance. ° = - (2 marks)
- c.' A thermistor with a resistance of $10\text{k}\Omega$ at 25°C has a temperature coefficient of resistance of 4% PC.
Calculate its resistance at 30°C . a (2 marks)
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Paper 3 of 3 — EEE 320 Measurement and Instrumentation

Level 300 • First Semester • PastQuestion

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COURSE CODE: EEE 320
 COURSE TITLE: Measurement and Instrumentation
 SESSION: 2017/2018
 DEGREE: B.ENG
 TIME ALLOWED: 3 Hours

INSTRUCTION: Answer part A and answer two question(s) from part B also answer two question(s) from part C

PART A

1.A

Using the Schering bridge in figure 1, calculate the values of the:

(i) Capacitance C_x and the equivalent series resistance R_x (5 ½ marks)
 (ii) Power factor and loss angle (2 ½ marks)

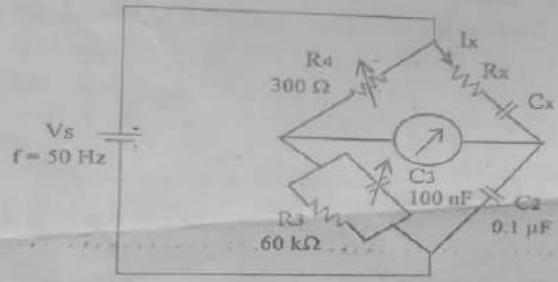


Figure 1

1. (iii) Explain the following measurement parameters: (i) Precision (ii) Reproducibility (iii) Resolution (3 marks)

1. (iv) Explain the types of faults that are most likely to occur in cables. (3 marks)

1B.

(i) Define the following terms: (i) Digital system (ii) Analog system. (2 marks)

(ii) A simple digital to analog conversion circuit (weighted resistor) has '1' signal indicating 'ON' and '0' indicating 'OFF' position. If the input signal ($D_3 D_2 D_1 D_0$) is given by 1101, find the analog output V_0 . $V_{ref} = -5V$. (3 marks)

(iii) State four features of an ideal strain gauge and three applications of strain gauges. (5 marks)

(iv) The reference-voltage of a dual slope ADC is 15 V. The integrator has a capacitor of 1 μF and a resistance of 200 kΩ. The time taken to read an unknown voltage V_x is 0.6 s. Find V_x . (2 marks)

(v) State the three laws of thermocouple. (3 marks)

PART B**Question 2**

2. (A) Two voltmeters are used to measure V_{AB} in a series circuit of two resistors $R_1 = 30 \text{ k}\Omega$ and $R_2 = 20 \text{ k}\Omega$ supplied by a 30 V voltage supply. The sensitivity of meter 1 is $1 \text{ k}\Omega/\text{V}$ and the sensitivity of meter 2 is $20 \text{ k}\Omega/\text{V}$. Both meters are used on a 10 V range. Estimate how good meter 2 is compared to meter 1 in terms of % error. (5 marks)

2. (B) Explain the following:

(i) Ammeter insertion effect (2 marks)

(ii) Voltmeter loading effect (2 marks)

Question 3

3. (A) A rectifier ammeter with the circuit shown in Figure 2 is to give FSD for a primary current of 250 mA. The PMMC meter has FSD 1 mA and $R_m = 1700 \Omega$. The current transformer has $N_s = 500$ and $N_p = 4$. The diodes each have $V_F = 0.7 \text{ V}$, and the series resistance is $R_s = 20 \text{ k}\Omega$. Calculate the required value of R_L . (7 marks)

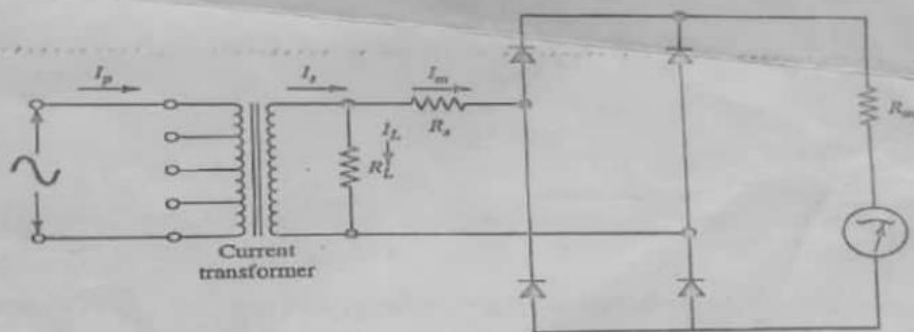


Figure 2

B. Outline the methods of locating cable faults (3 marks)

Question 4

4. (A) Explain why the Wheatstone bridge is unsuitable for measuring very low resistances. What bridge is preferred, and how does it compensate for the weakness of the Wheatstone bridge? Show relevant sketches. (5 marks)

(B). Find the value of the multiplier resistor required to convert a 50 μA meter into a voltmeter

having a full scale voltage of 5V. The resistance of the meter is 1800 ohms. Also, determine the sensitivity of the meter. (5 marks)

PART C

Question 5

- (a) Find the successive approximation of A/D output for a 4-bit converter to an 8.217V input if the reference is 5V. (4 marks)
- (b) (i) Determine the output voltages caused by each bit in an 8-bit ladder if the input levels are 0 = 0V and 1 = +16V. (ii) Also determine the resolution and full-scale output of this circuit. (iii) Find out the voltage from the above ladder for a digital input of 11100111. (6 marks)

Question 6

- (a) Briefly describe, with suitable diagram, the operation of a linear variable differential transformer (LVDT). (4 marks)
- (b) An electrical resistance strain gauge of resistance 150Ω and gauge factor 2.0 is bonded to steel having an elastic limit stress of 400 MN/m^2 , and modulus of elasticity 200 GN per m^2 . Determine the resistance change (i) due to change of stress equal to one-tenth of the elastic range and (ii) due to a change in temperature of 20°C , if the material is an advance alloy.

Use: Temperature coefficient resistance for advance alloy = $0.2 \times 10^{-4}/^\circ\text{C}$, temperature coefficient of linear expansion for advance alloy and steel are $0.2 \times 10^{-5}/^\circ\text{C}$ and $0.2 \times 10^{-5}/^\circ\text{C}$. (6 marks)

Question 7

- (a) Explain briefly the principle of operation of Liquid column gauge and McLeod gauge (4 marks)
- (b) The following table has been prepared from the calibration chart of iron-constantan thermocouple, with reference temperature at 0°C . Suppose the temperature of the hot junction is measured with an iron-constantan thermocouple, with the reference junction temperature at 40°C . If the voltage measured is 9.64 mV, find the actual temperature of the hot junction. (2 marks)

Temperature ($^\circ\text{C}$)	15	30	40	...	180	190	200	208	210
Emf (mV)	0.778	1.56	2.11	...	9.64	10.25	$\frac{10.7}{4}$	11.2	11.75

- (c) Briefly describe "Thermistors", showing their characteristic Negative and Positive (4 marks)

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL/ELECTRONICS ENGINEERING

EXAMINATION: Second Semester

COURSE CODE: EEE 320

COURSE TITLE: Measurement and Instrumentation



SESSION: 2018/2019

DEGREE: B.ENG

TIME ALLOWED: 3 Hours

INSTRUCTION: Answer part A and answer two question(s) from part B also answer two question(s) from part C

PART A

1.A

Using the Schering bridge in figure 1, calculate the values of the:

(i) Capacitance C_x and the equivalent series resistance R_x

(5 ½ marks)

(ii) Dissipation factor and loss angle

(2 ½ marks)

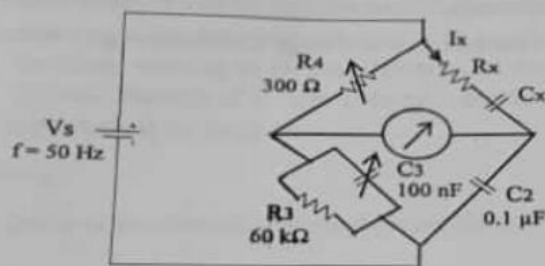


Figure 1

(iii) Explain the following measurement parameters: (i) Precision (ii) Reproducibility

(iii) Resolution

(3 marks)

(iv) Explain the types of faults that are most likely to occur in cables.

(3 marks)

1B.

(a) A simple digital to analog conversion circuit (weighted resistor) has '1' signal indicating 'ON' and '0' indicating 'OFF' position. If the input signal ($D_3 D_2 D_1 D_0$) is given by 1101, find the analog output V_{out} . Where, $V_{ref} = -5V$. (3 Marks)

(b) A strain gauge with gauge factor of 2 is fastened to a metallic member subjected to a stress of 1000 kg/cm^2 . The modulus of elasticity of the metal is $2 \times 10^4 \text{ kg/cm}^2$. Calculate the percentage change in the resistance of the strain gauge. What is the value of Poisson's ratio? (3 Marks)

(c) A copper resistor with resistance of 15Ω at 20°C is used to indicate the temperature of bearings of a machine. Determine the limiting value of resistance if the maximum

Page

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- temperature of the bearing does not exceed 175°C . The temperature coefficient of resistance of copper at 20°C is $0.00425\ \Omega/\Omega^{\circ}\text{C}$. (3 Marks)
- (d) An LVDT is used for measuring deflection of bellows. The sensitivity of LVDT is $40\ \text{V/mm}$. The bellows is deflected by $0.125\ \text{mm}$ by a pressure of $0.8 \times 10^6\ \text{N/m}^2$. Determine the sensitivity of the LVDT in V per N/m^2 and the pressure when the voltage of LVDT is $3.5\ \text{V}$. (3 Marks)
- (e) A 5-plate transducer has plates of dimensions $20\text{mm} \times 20\text{mm}$ and separated 0.25mm apart. The arrangement is to be used for measuring displacement. Determine the sensitivity of the arrangement. Assume air medium. ($\epsilon_0 = 8.854 \times 10^{-12}\ \text{F/m}$). (3 Marks)

PART B

Question 2

- (A) Two voltmeters are used to measure V_{RS} in a series circuit of two resistors $R_1 = 30\ \text{k}\Omega$ and $R_2 = 20\ \text{k}\Omega$ supplied by a $30\ \text{V}$ voltage supply. The sensitivity of meter 1 is $1\ \text{k}\Omega/\text{V}$ and the sensitivity of meter 2 is $20\ \text{k}\Omega/\text{V}$. Both meters are used on a $10\ \text{V}$ range. Examine how good meter 2 is compared to meter 1 in terms of % error. (5 marks)
- (B) Explain the following: (2 marks)
- Ammeter insertion effect (2 marks)
 - Voltmeter loading effect

Question 3

- (A) A rectifier ammeter with the circuit shown in Figure 2 is to give FSD for a primary current $250\ \text{mA}$. The PMMC meter has FSD $1\ \text{mA}$ and $R_m = 1700\ \Omega$. The current transformer has $N_s = 5$ and $N_p = 4$. The diodes each have $V_F = 0.7\ \text{V}$, and the series resistance is $R_s = 20\ \text{k}\Omega$. Calculate required value of R_L . (7 marks)

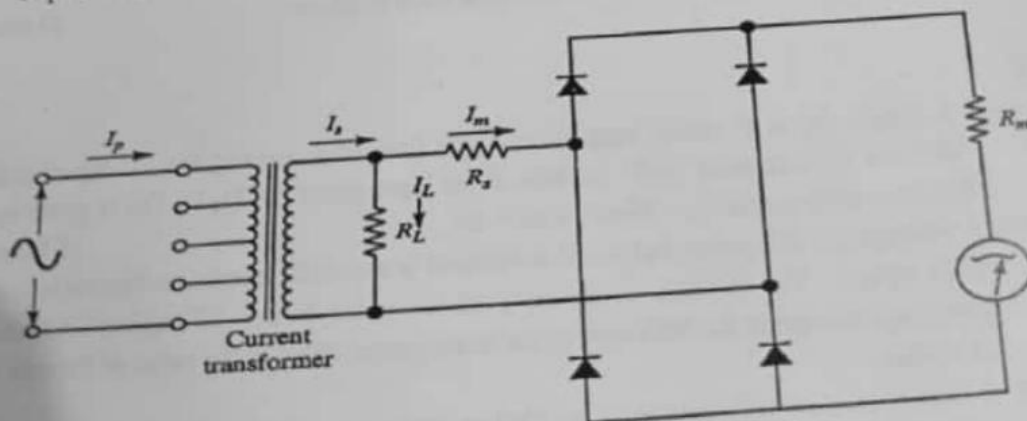


Figure 2

B. Outline the methods of locating cable faults

(3 marks)

Question 4

(A) Explain why the Wheatstone bridge is unsuitable for measuring very low resistances. What bridge is preferred, and how does it compensate for the weakness of the Wheatstone bridge? Show relevant sketches. (5 marks)

(B). Find the value of the multiplier resistor required to convert a 50 μ A meter into a voltmeter having a full scale voltage of 5V. The resistance of the meter is 1800 ohms. Also, determine the sensitivity of the meter. (5 marks)

PART C

Question 5

(a) State the 3 Laws governing thermocouples. (4 Marks)

(b) A capacitive transducer is made up of two concentric cylindrical electrodes. The length of the electrodes is 25 mm, the inner diameter of outer cylindrical electrode is 4.2 mm and the outer diameter of the inner cylindrical electrode is 4.0 mm. Determine the sensitivity of the transducer, assuming air medium. Determine the change in capacitance for a displacement of the inner electrode of 2.5 mm. Determine also the electric stress when voltage of 150V is applied across the electrodes. (6 Marks)

Question 6

(a) Briefly explain bimetallic strip and thermopiles as temperature measuring devices. (4 Marks)

(b) Determine the output voltages caused by each bit in a 6-bit ladder if the input levels are 0=0V and 1=+16V. Determine the resolution and full-scale output of this circuit. Find out the voltage from the above ladder for a digital input of 101011. (6 Marks)

Question 7

(a) Prove that for strain gauges having a conductor length l , area of cross-section A , resistance R and resistivity ρ , for small variations $\frac{\Delta R}{R} = \frac{\Delta L}{L} - \frac{\Delta A}{A} + \frac{\Delta \rho}{\rho}$. (4 Marks)

(b) Determine the thermoelectric sensitivity and emf developed in a thermocouple made of copper and constantan for a temperature difference of 200°C between its junctions. Given that the thermoelectric emfs of copper and constantan against platinum are 7.4 and -34.4 μ V per °C temperature difference. State the law of thermocouple that was used in solving the question. (6 Marks)

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